

**DERIVATION OF DESIGN DATA IN
COMPOSITES**

Part: **1.4 – COMPOSITE MATERIALS**

Chapter: **1480**

Title: **DERIVATION OF DESIGN DATA IN
COMPOSITES**

Revision: **1.1**

Date: **27/10/2021**

1 INTRODUCTION

1.1 OBJECT AND SCOPE

This chapter discusses the general considerations to take into account for deriving design values of composite materials from data available from mechanical testing. Further explanations shall be given in following chapters of the Stress Methods Manual for each specific type of composite material (fibrous composite, sandwich core, adhesive, ...).

Initial remarks:

- ✓ Only materials fully qualified according to authorized specifications for aerospace applications will be allowed to be used in a part with any structural responsibility. For composite materials, this category includes only Airbus specifications.
- ✓ Materials Design Values to be used shall be consistent with applicable Stress Methods. The material datasheets included in this manual are authorized sources of design data. Contact the Stress Methods function for further advice.

Some contents of this chapter were extracted from Ref. 3 (scope restricted to design office at Airbus DS – Getafe site) and later discussed and adapted with Airbus DS harmonization process.

2 STATISTICAL ANALYSIS

The generation of material design values/allowables for structural analysis, at coupon and structural element levels of the pyramid of tests, shall require the use of statistical analysis methodologies to derive the basis-values requested by regulations:

- ✓ **A-basis Value** -- A statistically-based material property; a 95% lower confidence bound on the first percentile of a specified population of measurements. Also a 95% lower tolerance bound for the upper 99% of a specified population

DERIVATION OF DESIGN DATA IN COMPOSITES

- ✓ **B-basis Value** -- A statistically-based material property; a 95% lower confidence bound on the tenth percentile of a specified population of measurements. Also a 95% lower tolerance bound for the upper 90% of a specified population. Therefore it is a value that at least 90% of the population will exceed with a 95% confidence

The applicable statistical methods for calculation of basis values shall be those defined by the CMH-17, Volume 1, Chapter 8 "STATISTICAL METHODS" and, for small data sets, those defined per Ref. 2 (see following sub-sections). Note that only test specimens showing valid failure modes shall be used for determination of design values.

The selection of A or B basis values for use in structural analysis is linked to the notion of multiple or single load path. Traditionally within Airbus, composite structures are designed to be damage tolerant, with multiple load design principles and validations. For these reasons, for composite materials, B-basis values are used as allowable design values.

Design values for mechanical properties shall be presented in terms of mean values and B-basis values (one scalar value per basis and ambient condition), and its ratio K_B (which shall be expected to be a unique number per property after pooling data across environmental conditions).

2.1 MEAN OR BASIS VALUES?

Rules about the kind of statistical value of a mechanical property that shall be used in analyses with composite materials:

- ✓ Mean values of elastic moduli shall be used for internal loads/stress calculations
- ✓ B-basis values of ultimate allowable stresses/strains shall be used for strength analyses

In buckling stability analyses the B-basis values of the elastic moduli (in compression loads) shall have to be used

2.2 THE SMALL SAMPLE FORMULA

This method is used for the generation of composite design allowables based on small datasets (between 3 and 18 specimens). Per Ref. 2, the B-basis value for a small population must be calculated as follows:

$$V_B = \bar{V} \times \frac{1 - k_b \times C_v}{1 + Conf \times \frac{C_v}{\sqrt{n}}}$$

$\bar{V}$: Mean value
n: number of specimens
 $k_b = 1.2816$
Conf = 1.6449
Cv: coefficient of variation

Note: coefficient of variation for a given set of data

$$C_v = \frac{s}{\bar{V}}, \text{ where } s \text{ is the standard deviation, i.e. } s = \sqrt{\frac{1}{n-1} \cdot \sum_{i=1}^n (V_i - \bar{V})^2}$$

The value of Cv to be used shall be the maximum between the value calculated for the sample and the value known for the population (see following page)

DERIVATION OF DESIGN DATA IN COMPOSITES

The typical number of specimens per batch (n) is 6 for mechanical tests. At least 1 batch of 6 specimens should be tested per property/test-method and environmental condition.

The population coefficients of variation for pre-impregnated fibre reinforced plastics used are:

CV = 5.8 %	CV = 9.0 %	CV = 4.7 %
Lamina & Laminate: - Tensile strength for plain specimens. Laminate: - Open hole tensile strength. - Filled hole tensile strength.	Lamina: - Inter-laminar shear strength. - In plane shear strength. - Compressive strength for plain spec. Laminate: - Compressive strength for plain spec. - Open hole compressive strength. - Filled hole compressive strength. - Shear strength. - Bearing strength. - CAI, TAI, SAI.	Lamina & Laminate: - Modulus.

Table 1. *Coefficients of variation of populations for composite laminates*

- ✓ This hypothesis of similar data dispersions (same coefficients of variation), as function of the failure mode, has been checked for several pre-impregnated materials in the same reference, and it is considered also adequate for FRP materials manufactured by resin infusion processes.
- ✓ For adhesives, the resin-based CV of 9% as population Coefficient of Variation is proposed to derive the design B-basis allowable when using the SSF.
- ✓ For honeycomb cores, a value for the population Cv equal to 15% is proposed to derive the design B-basis allowable when using the SSF. This upper bound is consistent with most available data presented in material databases.

2.3 LARGER SAMPLES AND POOLING DATA

In case a minimum of 18 specimens is available for the given property the CMH-17 procedure in Vol. 1, Chapter 8 shall be applied. The MS Excel sheet *STAT17* (provided by the CMH-17 Secretariat) shall be used for this purpose.

Apart from the inherent reduction of uncertainties when more coupons are tested, and the natural improvement of the resulting B-basis value, this detailed analysis is capable to produce higher allowables, detecting when the results for different batches can be considered belonging to the same population, grouping properly those data, identifying potential outliers, or approximating the data by the most appropriate statistical estimator.

For pooling data across environments in composites data analysis, if some conditions are accomplished, a possible step beyond the single-point method implemented in *STAT17* for improving the basis design values shall be to apply either the *Pooled Standard Deviation* or the *Pooled Coefficient of Variation* methods (see details in CMH-17, Vol. 1, section §8.3, in particular §8.3.5, §8.3.10). The MS Excel sheet *ASAP* (provided by the *NIAR*) can be used for this purpose.

DERIVATION OF DESIGN DATA IN COMPOSITES

If deriving basis values from screening or qualification tests data of new materials, the recommended strategy shall be to apply to calculations the *Modified Coefficient of Variation* approach (CMH-17, Vol. 1, §8.4.4), in order to anticipate more realistic variability of material along future time.

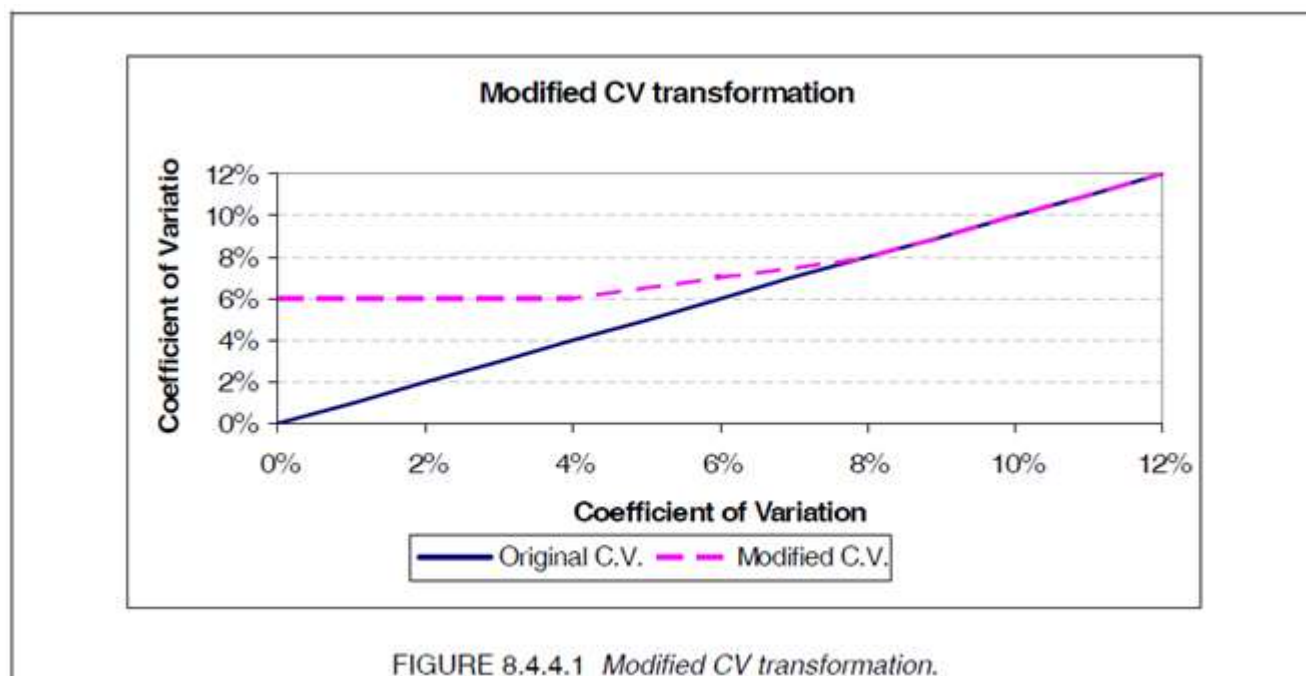


Figure 1. Modified CV approach as suggested in CMH-17

This later correction shall not be considered necessary if basis values are calculated from the legacy data available for a material already qualified in the past, especially if the input data used in the statistical analysis were obtained in tests programs separated in time (by years).

3 NORMALISATION RULES

Test instrumentation will normally be capable to obtain direct measurements of a limited number of physical magnitudes, as forces, strains, pressures or temperatures, but not, for example material stresses. It will be therefore necessary to perform some post-process, apart from the statistical analysis, on the test measurements, in order to obtain the magnitudes of interest. Some typical of fibrous composites, not exhaustive:

- ✓ Section stress: force sustained by a material section divided by the (gross or net) cross-sectional area (PT, PC, OHT, ILSS, ...)
- ✓ Elastic modulus: section stress (see above) divided by section strain
- ✓ Bearing stress: force sustained by a riveted joint divided by the bearing area (rivet diameter multiplied by the plate thickness)

DERIVATION OF DESIGN DATA IN COMPOSITES

As far as possible and up to certain extent, our test specifications shall be consistent with the available standard test methods (AITM, ASTM, ISO-EN, ...) which define not only the test coupons and fixture, but also (part of) the necessary post-process of the results data.

For generation of material design values for composite materials Airbus DS shall be consistent with rules and recommendations regarding normalization in Ref. 4, and regarding formatting (units systems, numbers rounding, ...) in Ref. 5.

Some rules normalization of mechanical properties shall be specific of the type of composite material (fibrous composites, sandwich cores,), and other intimately related to the stress methodologies applied later in the structural analyses. These rules shall not be discussed in this chapter. However some other rules are more generally applicable to different types of materials and, on the other hand, it is of major importance that are agreed with other functions participating in the activities of qualifying and testing composite materials.

- Thickness (t , t_{ply} , CPT)

For those materials which are provided in ply format, like fibrous composites and many adhesives, the (cured) ply thickness is a relevant parameter for design. In addition to actual/measured values observed in specimens and manufactured parts, a nominal value of the ply thickness shall be defined by the M&P function, which, in addition to other considerations, shall have to be representative of real production parts.

- In fibrous composite materials, properties dominated by the reinforcement fibres (and not significantly influenced by the matrix resin content) shall be normalized using the nominal value of the ply thickness.
- On properties dominated by the resin matrix, the transformation of tests forces into stresses across the resistant sections shall be done using the actual thickness measured in tests specimens.

- Bolt/rivet diameter (d)

In tests related with riveted/bolted joints, typically normalization will require dividing test loads by the rivet diameter. The standard rules will be:

- Use the diameter of the cylindrical shank (not any of the countersunk part)
- Use the nominal value (not a measured value)

An allowed exception to the rule above about using the bolt-shank diameter for normalizing strength is the case of the pull-out (or pull-through) laminate failure, when the rivet is loaded in tension: in that case normalizing the strength with the external diameter of the rivet-head (or nut) will be also allowed.

- Cross-section width (w)

For converting forces sustained by a cross-section into loads per unit-width or stresses, use always the actual measured width of the specimen's section.

- Environmental Knock-Down-Factors

Along with the basis values at applicable environmental conditions and the K_B ratio, the environmental knock down factors ($EKDF$ or K_{EKDF}) for a property, shall be presented as the

DERIVATION OF DESIGN DATA IN COMPOSITES

ratios of the property at an ambient condition divided by the reference value at Room-Temperature for the dry material (both as mean values).

- *Design environmental conditions, elastic moduli and allowable strains in fibrous composites*

Design values of mechanical properties shall be presented for all the relevant ambient conditions with data available, or those applicable for the particular airframe, from Cold/Dry, RT/Dry up the Hot/Wet.

Anyway, some mechanical properties relevant for design shall have to be obtained in an indirect manner, normalized by another property which can show some sensitivity to environment. We are talking specifically about the strengths of fibrous composites expressed in terms of allowable strains, which are obtained by dividing ultimate stresses by the elastic modulus.

Normally, the E-KDF of the Young's modulus in fibre direction, within the material service range, is very close to 1.0. The design value for E_1 shall be set constant, non-dependant on service temperature (by extension, the same rule shall apply for E_2 in bi-axial fabric materials).

This modulus is the main driver of the stiffness of any laminate built following the standard design rules (minimum 10% fiber at each of the reference directions 0° , 90° , 45° and -45°). Thus, an alternative to the E-KDF approach for the elastic moduli, is to set the design values for ply moduli (all of them) as constant parameters applicable to any environmental condition within the usability range, valid for laminates following those standard design rules. This later approach shall be adopted by Airbus DS, using H/W data at a moderately high service temperature (typ. 70°C) to set the resin-driven moduli, and adjusting the fibre-driven moduli (to be explained in following chapter). These design values for the elastic moduli, treated as material-invariants, shall be used to:

- Transform material ultimate stresses into allowable strains when required,
- Calculate the stiffness of the composite parts and the internal loads distributions in stress analyses (*),

... across all service environments (load cases and associated loads will be associated to different ambient conditions).

(*) However, in certain analysis scenarios, as composite-metal hybrid assemblies, alternative approaches may be preferred to account properly thermal loads (as setting the design moduli at the mid-point of the temperature increment/decrement of the load case).

Considerations beyond discussion above and the pointed references shall be treated specifically in following chapters, each of them dedicated to a different type of composite material.

DERIVATION OF DESIGN DATA IN COMPOSITES

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
AVG , $\bar{V}$	?	Mean/average value of a set of data (measurements of a property in a test batch)
CPT	mm	Cured Ply Thickness
C_v , CV	%	Coefficient of variation of a set of data or a population of samples (STD/AVG)
E-KDF, K_{E-KDF}	-	Environmental Knock Down Factor: ratio between the value of a mechanical property obtained at H/W conditions and the corresponding value at RT/Dry conditions
k_b	-	In lower-case, B-basis "tolerance factor" as defined per the CMH-17, and related to the coefficient of variation CV and the B-basis knock down K_B factor as follows: $V_B / \bar{V} = K_B = 1 - k_b \cdot CV$
K_B	-	In upper-case, ratio between the B-basis value and the mean-value of a data sample or population
M&P	-	Materials & Processes function in Airbus Defence and Space
n	-	Number of coupons in a single batch (typically 6 or more)
s , STD	?	Standard deviation of a set of data
SSF	-	Small Sample Formula
t	mm	Thickness (time in sub-chapter about dimensional analysis)
T	°C	Temperature
V_B	?	B-basis value: lower bound of a property for the 90% of the population with 95% confidence

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[FAA/SAE] CMH-17	03/12	COMPOSITE MATERIALS HANDBOOK
Ref. 2	[Airbus] PP0300181	2	Composite statistical method to determine B-basis value from small data sets"
Ref. 3	TAE-T-MM-170009	Is. C	GUIDELINES FOR RESULTS ANALYSIS OF COUPONS TESTS IN COMPOSITE MATERIALS
Ref. 4	[Airbus] RP0412191	1	Normalization rules for composite properties
Ref. 5	TEA-T-MM-190002	Rev. B	GUIDELINES FOR PRESENTATION OF RESULTS OF COUPONS TESTS IN COMPOSITE MATERIALS

DERIVATION OF DESIGN DATA IN COMPOSITES

SOFTWARE

Program	Version	Description
STAT17 (FAA/SAE CMH-17)	Rev. 3 2001	Statistical analysis of composite materials tests data by the single-point method
ASAP (NIAR - Wichita State University)	2004	AGATE Statistical Analysis Program/Procedure for pooling across levels of a fixed effect
RECIPE (NIST)	1.0 1995	One-sided tolerance limits (in particular, A- and B-basis material property values) for regression models in the presence of between-batch variability

Table 2. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 28/07/2020	G. Baños	First issue	All pages
1.1 06/06/2021	G. Baños	Conditions for application of the modified coefficient of variation approach in calculation of basis values	Pages 3-4

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 06/06/2021	F. Glock 16/06/2021
Materials & Processes (TEPMO)	-	-

Table 4. *Approval signatures table for last revision of this chapter*